

BRILLIANTA ZAYYAN MUHAMMAD

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PROFILE

An Informatics Engineering graduate specializing in Artificial Intelligence implementation, focusing on Generative AI, Large Language Models (LLMs), Multi-Agent architectures, and Computer Vision. Experienced in architecting end-to-end AI systems, including RAG-based chatbots, autonomous research agents, and automated video processing pipelines, supported by practical skills in AI backend development using FastAPI and Docker. Complementing technical expertise with a background in IT support and hardware/software troubleshooting to ensure seamless technical operations. Committed to continuous innovation, actively leveraging advanced prompt engineering and AI-driven automation to optimize development workflows. Currently seeking opportunities to develop and deliver real-world intelligent solutions as a AI/ML Engineer.

EDUCATION

STATE UNIVERSITY OF MALANG

Bachelor of Engineering

Major in Informatics Engineering / Computer Science

Cumulative GPA: 3.93/4.0

Relevant Coursework: Machine Learning, Data Science, Computer Vision, Database & Information Systems

Malang, Indonesia

Graduated 2025

COMPETITION

INDONESIA INVESTORS DAY IN BALI (BRONZE AWARD/3RD PLACE)

August 2024

- Ideated the concept for a health-related mobile application, which became the foundation of the project.
- Served as Machine Learning Engineer, developing core ML functionalities.
- Assisted the mobile development team as a secondary role.
- Collaborated effectively within a multidisciplinary team to deliver an award-winning solution.

SATU DATA INFOGRAPHICS (PARTICIPATED)

November 2023

- Coordinated as a team leader.
- Implemented data understanding on the given problem.
- Discovered insights from the problem, visualizing them using Tableau and presenting them as infographics.

EXPERIENCE

AI ENGINEER AT PT LINK APISINDO MEDIA (INTERNSHIP)

October 2025 – April 2026

- Developed end-to-end AI systems, including RAG-based chatbots, an autonomous research agent, and automated video processing pipelines.
- Designed scalable backend architectures using FastAPI, LangGraph, and Redis, focusing on real-time data synchronization and asynchronous task management.
- Implemented LLM evaluation and monitoring workflows using LangWatch and LangSmith to ensure response accuracy, track latency, and optimize model reliability.
- Leveraged AWS Bedrock, DeepSeek, and Gemini to deliver high accuracy solutions for legal, information, and creative sectors.
- Configured the initial OpenClaw setup and developed a Model Context Protocol (MCP) to integrate ICCN RAG knowledge into the OpenClaw ecosystem, enabling intelligent chatbot interactions via Telegram Bot.

COMPUTER VISION RESEARCH ASSISTANT

October 2025 – November 2025

- Assisted in managing and preparing instance segmentation datasets using Roboflow, and conducted experiments with multiple YOLOv8-seg variants.
- Contributed research insights and recommendations throughout the study on the impact of various data augmentation combinations on model performance.

WEB DEVELOPER AT DISHUB KOTA MALANG (INTERNSHIP)

July 2024 - August 2024

- Developed a queue management website enabling visitors to monitor live queue status.
- Built an internal web application to assist staff in generating reports more efficiently.

MACHINE LEARNING COHORT AT BANGKIT

February 2024 - June 2024

- Completed courses on Machine Learning delivered through Coursera and Dicoding platforms.
- Collaborated in a multidisciplinary capstone team to develop a mobile application addressing health-related issues.

PROJECTS

ICCN AI AGENT IMPLEMENTATION

January 2026 – April 2026

- AI Research Agent : Developed a comprehensive search engine utilizing LangGraph for multi-step reasoning and the ReAct (Reason & Action) paradigm, integrated with Tavily for advanced web search. Implemented automated Markdown report synthesis with accurate citations across multi-model LLMs.
- Ramalan Agent (Gamification Data Collector): Created a gamified task-oriented agent for user profiling, utilizing Redis for robust session management across multi-turn conversations. Optimized UI responsiveness by delegating heavy LLM computations to asynchronous background tasks.

JDIH JAWA TIMUR REAL-TIME CHATBOT

January 2026 – March 2026

- Developed a RAG-based legal information assistant using FastAPI and ChromaDB, implementing Hybrid Search (Semantic & Lexical) with Reciprocal Rank Fusion (RRF).
- Built a real-time data ingestion pipeline to generate document embeddings and synchronize MySQL CMS with a vector database (ChromaDB), reducing new legal document availability latency from ~24 hours to <1 minute.
- Optimized accuracy for complex legal document numbering through dynamic Regex Extraction within the lexical search framework.

AI CLIPPER

February 2026 – March 2026

- Designed an end-to-end pipeline to extract viral clips from YouTube or Local Videos using Faster-Whisper for transcription and Amazon Bedrock model for semantic analysis.
- Developed a “Smart Crop” feature using OpenCV face detection to automatically reframe landscape videos (16:9) into vertical formats (9:16).

KLINIKHOAX AI IMPLEMENTATION

October 2025 – December 2025

- Developed the "KlinikHoax" chatbot using LangFlow and Amazon Nova, creating an end-to-end system for verifying news and tracking user tickets.
- Engineered an AI Fact-Check feature that cross-references user reports with real-time Google Search results, significantly reducing manual verification time for admins.

BISINDO SIGN LANGUAGE DETECTION USING YOLO (THESIS)

August 2024 – June 2025

- Built an instance segmentation model using YOLOv8 to detect BISINDO alphabet gestures.
- Conducted data preparation through collection, annotation, cleaning, and augmentation using Roboflow.
- Evaluated and compared model performance across different data split scenarios.

FITFOOD IMAGE CLASSIFICATION

August 2024

- Trained a multi-label deep learning model to detect and classify 40+ traditional Indonesian food items using Transfer Learning with an InceptionV3 architecture.
- Fine-tuned 60+ top layers to optimize accuracy and built an efficient TensorFlow data pipeline for image processing.
- Converted the model to a lightweight TensorFlow Lite (.tflite) format for Android deployment and assisted the mobile team with frontend integration.

ROGUE OF LOGIC GAME

December 2023

- Implemented high-quality and immersive sound effects that enhance the gaming experience.
- Used programming languages and tools to code the UI based on the design specifications.

CERTIFICATIONS AND TRAINING

- Bangkit Academy 2024 led by Google, Tokopedia, Gojek, & Traveloka
- ICCN Contribution Appreciation Certificate
- DeepLearning.AI TensorFlow Developer Specialization
- TensorFlow: Advanced Techniques

- TensorFlow: Data and Deployment
- Generative AI for Everyone
- Machine Learning Specialization
- Learn Data Analysis with Python (Dicoding)
- Additional certifications focused on foundational programming, Python, Git, data visualization, and mathematics for machine learning, including courses from Dicoding, Coursera (Google & IBM), and other platforms.

PUBLICATION

Muhammad, B. Z., & Rosyid, H. A. (2022). Peran dan Tantangan Kecerdasan Buatan dalam Mengembangkan Smart Stadium menuju Era Society 5.0. Jurnal Inovasi Teknik dan Edukasi Teknologi, 2(10), 452-456. DOI:10.17977/um068v1i102022p452-456

SKILLS

Languages : Fluent in Indonesian (Native), Proficient in English (reading & writing)

Programming & Databases : Python, SQL (MySQL, PostgreSQL), ChromaDB, Redis

AI & Frameworks : Generative AI (LLMs, RAG, Embeddings, Agents), Prompt Engineering, Computer Vision, Data Annotation, LangGraph, LangChain, LangFlow, FastAPI, FastMCP, YOLO, TensorFlow

Tools & Platforms : Docker, AWS Bedrock, Google Gemini, DeepSeek, Tavily, Faster-Whisper, LangSmith, LangWatch, Roboflow, MCP

Others : Typing Speed: ± 110 WPM (High Accuracy), IT Hardware/Software Troubleshooting